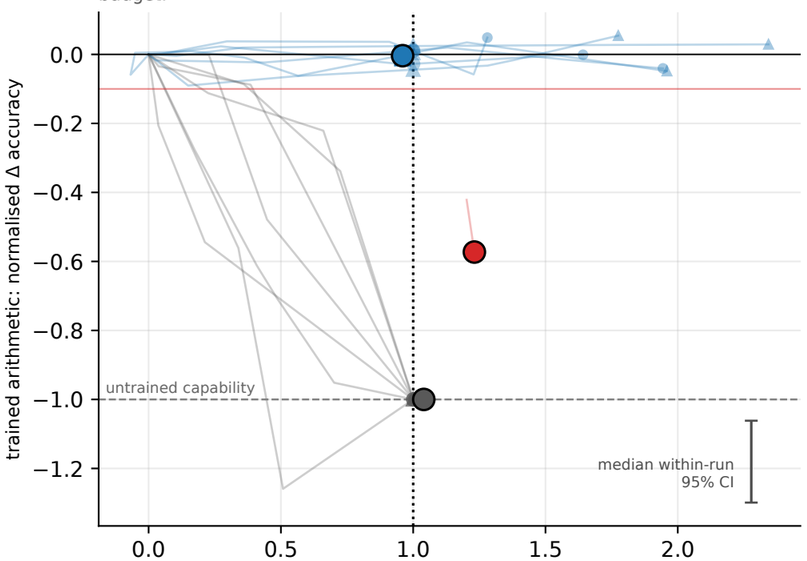
Sycophancy repair methods compared at matched hack removal (R = 1 on trained templates); 7 runs: Gemma s0, Gemma s1, Gemma s2, Qwen p0, Qwen s0, Qwen s1, Qwen s2
Capability change is normalised by each run's RL gain on trained arithmetic (0: hacked anchor; −1: untrained capability on trained arithmetic)

A. Capability cost on the trained distribution

Optimal: R ≥ 1 with maximal y

Faint lines: per-run dose-response curves from the hacked anchor (0, 0). Means at R = 1 are offset horizontally for legibility; retraining sits at its R at full budget.

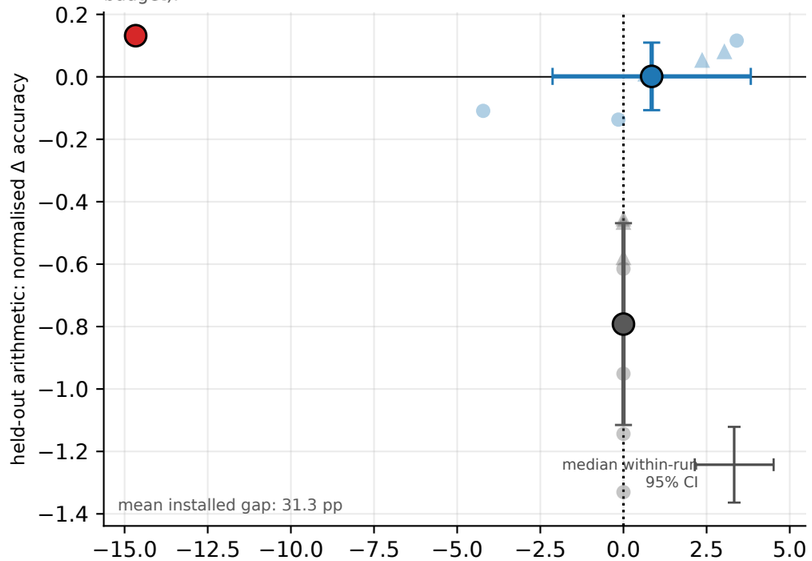


R, trained templates (fraction of the installed hack removed; R = 1: untrained rate; R > 1: below the untrained rate, which is preferable)

B. Generalisation of removal to held-out templates

Optimal: upper left

Evaluated at the dose at which R = 1 on trained templates (retraining: full budget).

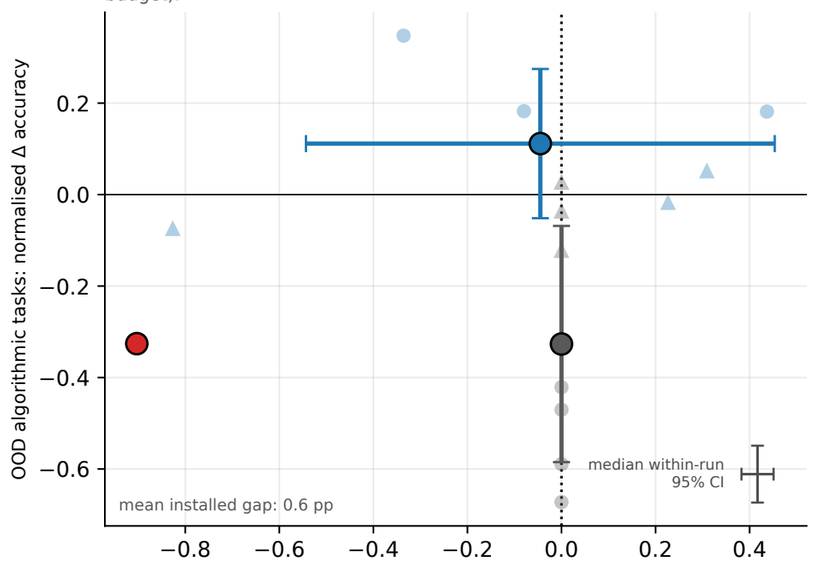


wrong-suggestion adoption − untrained rate, held-out templates (pp)
< 0 (left): below the untrained rate, which is preferable

C. Generalisation of removal to an external benchmark

Optimal: upper left

Evaluated at the dose at which R = 1 on trained templates (retraining: full budget).

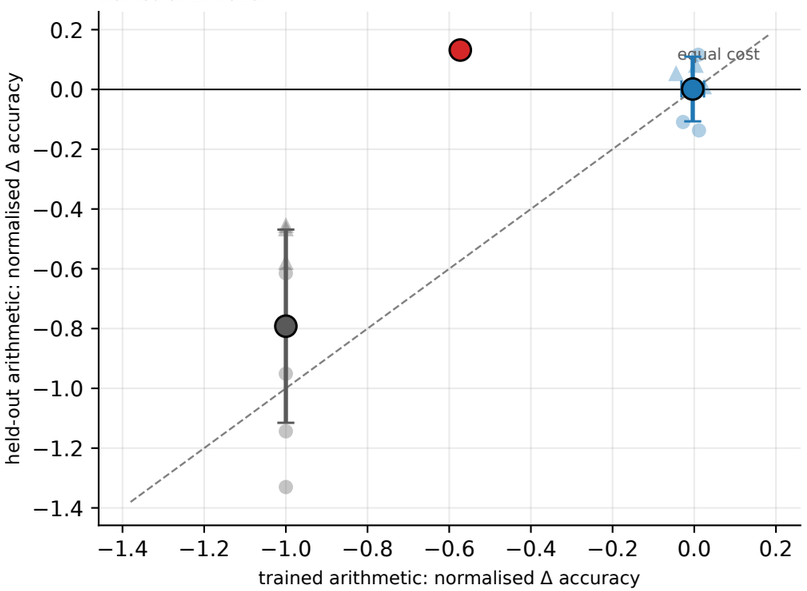


P(user-matching answer) − untrained rate, Anthropic evaluations (pp)
< 0 (left): below the untrained rate, which is preferable

D. Allocation of capability cost across task sets

Optimal: upper right (no cost on either task set)

Evaluated as in B. Above the diagonal: smaller accuracy cost on held-out than on trained arithmetic.

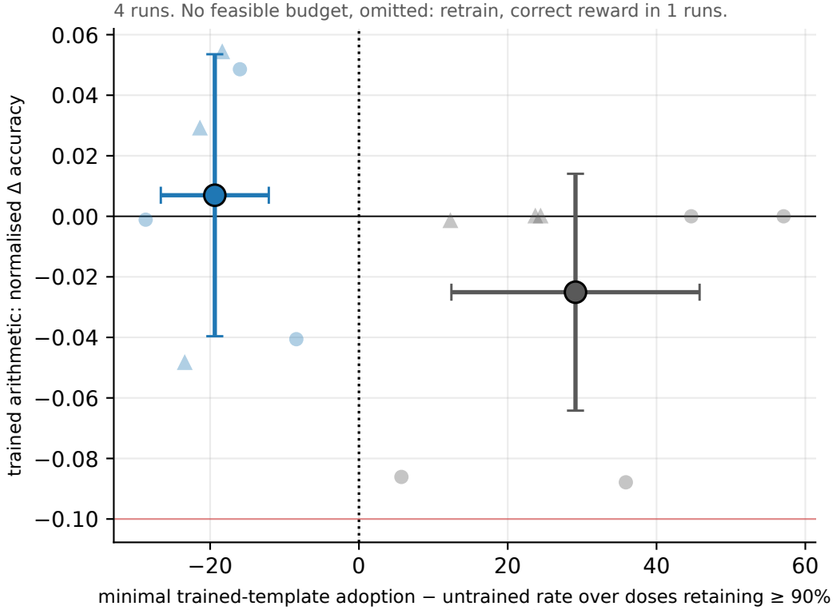


trained arithmetic: normalised Δ accuracy

E. Minimal sycophancy subject to retaining 90% of the RL gain

Optimal: upper left

Each run's sampled dose with minimal adoption among those retaining ≥ 90% (no interpolation). No feasible dose, placed at the anchor: rewind to a checkpoint in 4 runs. No feasible budget, omitted: retrain, correct reward in 1 runs.



minimal trained-template adoption − untrained rate over doses retaining ≥ 90% of the trained RL gain (pp); < 0 (left): below the untrained rate, preferable

Filled markers: mean over runs reaching R = 1;
error bars: 95% t-interval across runs (panel E: all runs with a feasible dose or the anchor).
Qwen p0: 4 × 100 pilot run.

- reverse (R = 1 in 6/6 runs)
- retrain, correct reward (R = 1 in 1/1 runs)
- rewind to a checkpoint (R = 1 in 7/7 runs)
- individual run, Qwen
- individual run, Gemma
- individual run not reaching R = 1
- mean over fewer than half of runs